FHS Formula Derivation — Basis Table

For: Adviser Validation (Ms. Ellen F. Mangaoang, DIT)
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Table 1. Theoretical and Literature Basis of the Financial Health Score (FHS) Formula

The SmartSpend Financial Health Score (FHS) is formally positioned as a **Prototype Observed Financial Health Indicator (FHI)**. It was not invented from scratch. Each design decision — the 0–100 scale, the equal-weight structure, the four components, and the two adjustment mechanisms — is grounded in an existing academic or institutional framework, adapted for the Philippine mobile application context.

FHS Design Element	What It Is in SmartSpend	Academic / Institutional Basis	How We Adapted It
Transaction-based scoring (0–100)	The FHS is computed entirely from the user's logged expense, income, and budget data — no surveys, no self-report	CBA–Melbourne Institute Observed Financial Health Indicator (Comerton-Forde et al., 2018, Technical Report No. 1) — uses actual bank-record data to derive a 0–100 observed scale	We adapted the transaction-based, observed-data approach to a mobile SQLite database instead of bank records
Equal-weight components	Each of the 4 FHS components is worth exactly 25 points	CBA–Melbourne Institute (2018); UNSGSA (Gubbins, Mazzotta & Rhyne, 2021) — endorses equal-weight design when longitudinal data is unavailable; **OECD/JRC Handbook on Constructing Composite Indicators (2008)** — formally establishes equal weighting as the methodologically defensible default	We used equal weights because we do not have multi-year behavioral data needed to justify differential weights for this population
"Prototype Observed FHI" naming	We call it a Prototype Observed FHI, not a Financial Health Score	Melbourne Institute Terms of Use require modified scales to use a different name. UNSGSA (2021) distinguishes Observed FHIs (transaction-derived) from subjective survey-based scales	We renamed to comply with Melbourne Institute guidance and distinguish from the CFPB Financial Well-Being Scale
Component 1: Savings Rate (25 pts)	Formula: `25 × min(1, savingsRate / 0.20)` Full score when user saves ≥20% of income. *No-income fallback:* 20 pts if spent < ₱5K; 15 pts if < ₱10K; 10 pts otherwise	Warren & Tyagi (2004) — the 50/30/20 rule establishes 20% savings as the standard personal finance target; FinHealth Score® (FHN) — Spend pillar includes savings rate as a core health indicator; UNSGSA (2021) justifies the fallback for informal-economy users	We used the 20% threshold directly from the 50/30/20 framework. No-income fallback uses spending volume as proxy
Component 2: Overspend Control (25 pts)	Formula: `25 × (1 − (hardOverDays + softOverDays × 0.5) / activeDays)` Soft overspend (single large purchase) = 0.5×; hard overspend (habitual daily) = 1×	Kahneman & Tversky (1979) — Prospect Theory: losses feel ~2× more painful than equivalent gains, making overspend penalties motivationally salient; Thaler & Sunstein (2008) — Nudge Theory: planned vs. habitual spending distinction	The soft/hard split prevents penalizing a one-time tuition payment the same as daily habitual overspending. 0.5× is a calibrated design constant
Component 3: Budget Adherence (25 pts)	Formula: `25 × (onBudgetCategories / totalBudgetCategories)` If no budgets set: auto-award 25 pts	FinHealth Score® (Financial Health Network, 2021) — Plan pillar includes budget adherence as a core health behavior; BSP CFIS (2025) — fewer than 50% of Filipino households maintain any written budget	Auto-award 25 pts when no budgets configured — does not penalize users for lack of initial setup

Component 4: Logging Consistency (25 pts)	Formula: `25 × (loggedDays / activeDays)` Full score when expenses are logged every active day	Deci & Ryan (2000) — Self-Determination Theory (SDT): Competence dimension — consistent behavior drives long-term habit formation; Ariely (2008); Duhigg (2012) — consistency in tracking is more predictive than volume on a single day	Tied directly to the Habit Streak feature. Streak milestone celebrations reinforce daily logging via SDT Competence
Warning Decay (−5 pts/day, max −15 pts)	If user continues spending in a budget category after an alert: FHS loses 5 pts/day for up to 3 days. Exempt: Bills, Health, Education	Kahneman & Tversky (1979) — Prospect Theory: loss salience creates a visible consequence, designed to increase the perceived cost of ignoring warnings. −5/3-day values are calibrated design constants	Decay only applies to discretionary categories. Non-discretionary obligations (bills, medical, tuition) are exempt
Lightweight Mode (4 alternate components when income not set)	Savings Rate → Spending Restraint; Budget Adherence → Category Balance; Logging Consistency stays; adds Habit Streak	UNSGSA (2021) — recommends transaction-derived indicators as an early-stage complement when full income data is unavailable	2 of 4 components redesigned for income-independent measurement so the app remains useful for students and informal-economy users
Lightweight: Spending Restraint (25 pts)	Formula: `ratio = spent/limit; 25×(1−((ratio−0.80)/1.20))` [clamped 0–25]. Full score at ≤80% of limit; 0 pts at ≥2× limit. *No-limit fallback:* `wantRatio = wantSpend/totalSpend; 10 + 15×(1−((wantRatio−0.30)/0.40))` [floor 10 pts]	Kahneman & Tversky (1979) — want vs. need spending distinction as behavioral health signal. 80% grace zone = personal finance buffer principle. Floor of 10 pts reflects BSP CFIS (2025) finding that most informal-economy users do not maintain formal budgets	Grace zone prevents penalizing users for being close to (but under) their limit. Floor prevents total score collapse when no limit is set
Lightweight: Category Balance (25 pts)	Formula: `topRatio = topCategory/discretionaryTotal` (Bills, Health, Education excluded); `25×(1−((topRatio−0.40)/0.60))` [clamped 0–25]. Full 25 pts when top category ≤40% of discretionary; 0 pts when one category = 100%	FinHealth Score® Spend pillar (FHN, 2021). Exempt categories follow BSP CFIS (2025) — Bills/Health/Education are non-discretionary fixed obligations. 40% threshold is a calibrated design constant	Denominator is discretionaryTotal (not total outflow) — rent and tuition legitimately dominate spending without indicating poor financial health
Lightweight: Habit Streak (25 pts)	Formula: `25 × min(1, currentStreak / 14)` Full 25 pts at 14-day consecutive logging streak	Deci & Ryan (2000) SDT Competence dimension; Lally et al. (2010) — consistent behavioral windows of ~14 days anchor initial habit formation routines (mean ~66 days; 14 days represents the earliest reliable phase)	14-day target is a calibrated design choice: long enough to build initial consistency, short enough to be achievable by students and new users

Gap Adjustment (+2/−3 pts/day)	+2 pts/day (max +10) for catch-up logging; −3 pts/day (max −15) for multi-day gaps	Ariely (2008); Duhigg (2012) — consistency in tracking is more predictive of financial health than volume on a single day. Asymmetry (penalty > bonus) follows Prospect Theory loss > gain principle	Capped at +10/−15 to prevent gaming (logging 100 expenses in one day) while still rewarding recovery after gaps. Values are calibrated design constants

Table 2. Alignment with Established Financial Health Frameworks

SmartSpend FHS Component	CBA–Melbourne Institute (2018)	FinHealth Score® (FHN)	CFPB Financial Well-Being Scale (2017)
Savings Rate	Observed savings behavior from bank records	Save pillar: savings rate and buffer	Q7: "I could handle a major unexpected expense"
Overspend Control	Observed overspending days (weighted: hard=1×, soft=0.5×)	Spend pillar: spending less than income	Q2: "My finances control my life" (reverse)
Budget Adherence	Category-level spending vs. limits	Plan pillar: on track with financial goals	Q9: "My finances are on track"
Logging Consistency	Data completeness metric	(Not directly measured)	(Not directly measured)
Warning Decay	(Not in framework — SmartSpend addition)	(Not in framework)	(Not in framework)
LW: Spending Restraint	(Not in framework)	Spend pillar: spending vs. limit	(Not directly measured)
LW: Category Balance	(Not in framework)	Spend pillar: spending distribution	(Not directly measured)

Key difference: The CFPB scale is self-reported (users rate their own financial feelings). The Melbourne Institute and FinHealth Score use transaction data — like SmartSpend. Our FHS reduces self-reporting bias on spending amounts; category attribution and logging completeness remain user-influenced.

Note on Category Balance formula: *The denominator is* `discretionaryTotal` *(Bills, Health, Education excluded) — not total outflow. Fixed obligations are excluded because BSP CFIS (2025) confirms these legitimately dominate Filipino household budgets.*

Table 3. Why Each Component Was Chosen — Linked to Needs Assessment Findings

Needs Assessment Finding	FHS Component That Addresses It	Survey Question That Measured It
Filipino respondents find it difficult to save (difficulty setting aside savings)	**Savings Rate** — rewards users who save ≥20% of income	Q11: "Do you set aside money for savings before spending?"
Respondents regularly exceed their planned budget	**Overspend Control** — penalizes days of habitual overspending	Q14: "How often do your expenses exceed your budget?"
Fewer than half of respondents maintain any budget	**Budget Adherence** — incentivizes setting and following category budgets	Q8: "Do you prepare a budget?"
Respondents track expenses inconsistently or not at all	**Logging Consistency** — full score only when logging every active day	Q21: "Do you currently record your expenses?" Q23: "How soon after purchase do you record?"

Note on Calibrated Numeric Constants

> The following specific numeric constants are **design calibrations** informed by the referenced theoretical frameworks but not directly derived from any single paper. They represent reasonable prototype values subject to empirical calibration in future longitudinal studies.

Constant	Value	Theoretical Basis	Design Rationale
Warning Decay rate	−5 pts/day	Prospect Theory (Kahneman & Tversky, 1979) — loss salience	Calibrated to be felt but not catastrophic in a 30-day cycle
Warning Decay cap	3 days / −15 pts max	Prospect Theory	Cap prevents total wipe-out from a single budget breach
Gap Adjustment bonus	+2 pts/day (max +10)	Consistency principle (Ariely, 2008; Duhigg, 2012)	Smaller than penalty to match loss-aversion asymmetry
Gap Adjustment penalty	−3 pts/day (max −15)	Prospect Theory asymmetry	Penalty > reward mirrors loss > gain finding
Soft overspend weight	0.5×	Prospect Theory / Nudge Theory — planned vs. habitual distinction	Single large purchases (tuition, medical) differ from habitual daily excess
Category Balance threshold	40% of discretionary	Concentration risk design principle	Above 40%, spending concentration becomes behaviorally relevant
Spending Restraint grace zone	80% of limit	Personal finance buffer principle	Remaining 20% provides a behavioral safety buffer before penalty begins
Habit Streak target	14 days	SDT Competence (Deci & Ryan, 2000); Lally et al. (2010) habit formation research	14 days anchors initial habit formation routines; achievable for students and new users

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Prepared to support adviser validation of the FHS formula derivation. All citations are verifiable via DOI or official publication page. Numeric constants disclosed as calibrated design values.